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Ethnomedicinal plants of Khunti district of Jharkhand state (India)

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Abstract : Khunti is a newly formed district of Jharkhand created by separating out from Ranchi. Majority of the population is tribes and indigenous people who live close to nature with their rich cultural heritage. The area is also very rich in plant biodiversity because yet not disturbed by human intervention. Being very close to capital city Ranchi shortly its natural green canopy will be disturbed. The cost of various developmental programmes and also the permanent settlers will be forced to leave their natural home. Therefore, there is an urgent need to take up ethnobotanical studies in general and ethnomedicinal studies in particular before it is too late. This type of study will give new impetus to the traditional system of healthcare. In view of the importance of traditional medicine which provides health services to large population, increased demand of herbal drugs by the pharmaceuticals and depleting plant resources, it is high time to document the medicinal utility of plants available in remote areas of the district. In the present paper, total of 40 plant species belonging to 26 families used to treat different types of diseases and disorders have been enlisted and their parts used have been focused.

Key Words: Ethnomedicine, traditional system, health care, herbal drug, indigenous.

INTRODUCTION

Ayurveda, Siddha, Unani and Folk (tribal) medicines are the major systems of indigenous medicines. Among these systems, Ayurveda is the most developed and widely practiced in India. Ayurvedic system depends heavily on herbal products. Several millions of Indian households have been using through the ages nearly 8000 species of medicinal plants for their health care needs of the total 45,000 species of wild plants. Over one and half million traditional healers use a wide range of medicinal plants for treating ailments of both humans and livestock across the length and breadth of the country. Over 800 medicinal plant species are currently in use by the Indian Herbal Industry. Unfortunately, our traditional system of medicines was cornered by the modern system of medicine, Allopath. Despite large numbers of benefits, the system has its own limitations. Moreover, it is very costly and shows many painful side effects.

Herbs are staging a comeback and herbal 'renaissance' is happening all over the globe. The herbal products today symbolise safety in contrast to the synthetics that are regarded as unsafe to human and environment. The revival of interest in the use and importance of medicinal plants has led to intensified efforts on the documentation of ethno- medicinal data of medicinal plants, since most traditional healers keep scanty records and their information is passed on mainly verbally, from generation to generation.[1] Research has been geared towards finding scientific evidence for the claims as to the therapeutic efficacy of herbs used by traditional healers. Most of the published and unpublished written ethnomedicinal data with valuable and complementary information are scattered in many documents, some of which are not easily available. An interdisciplinary systematization, which certainly helps to predict the most promising candidates for further laboratory or clinical investigations, appears as useful work.[1] Large numbers of

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infectious diseases are known to be treated with herbal remedies throughout the history of mankind. The maximum therapeutic and minimum side effects of herbal remedies have been verified in numerous scientific investigations. Even today, plant materials continue to play a major role in primary health care as therapeutic remedies in many developing countries.[2] It has been reported that natural products (their derivatives and analogs) represent over 50% of all drugs in clinical use, in which natural products derived from higher plants represent about 25% of the total. The World Health Organization estimated that over 80% of the people in developing countries rely on traditional remedies such as herbs for their daily needs and about 855 traditional medicines include crude plant extracts.[3] This means that about 3.5 to 4 billion of the global population rely on plants resources for drugs.[4]

In the present study, Khunti district has been selected for ethnomedicinal documentation. The Khunti district is a part of newly created state Jharkhand which has recently been carved out from the old Bihar. The area is dominated with different ethnic communities with their own customs and cultures. Despite being very close and bordering to capital city this region is not yet much disturbed and is abounded by dense green canopy. Moreover, large section of population relies on herbs and herbal medicines. It is visualized that shortly both forest cover and forest dwellers will be disturbed because of the increasing pressure of population and other types of load in the capital city Ranchi will be soon transferred to adjacent Khunti district. However, a few scattered information about ethnomedicinal studies are available from Khunti block but a comprehensive and systematic documentation of the whole area of Khunti district is lacking.

Therefore, the present investigation has been undertaken with objectives: to survey the area; to interact with local traditional healers and document their knowledge on medicinal plants; their usage and the types of diseases treated; and collection, identification of plants based on local names, so as to bring up to date ethnomedicinal data of this area.

STUDY AREA

Khunti district is one of the twenty-four districts in South Chotanagpur division of the Indian state of Jharkhand. The district of Khunti was carved out of Ranchi district on the 12th September 2007.The district has total surface area of 2610.86 Sq km. and Khunti town is the headquarters. It has six blocks: Arki, Karra, Khunti, Murhu, Rania and Torpa. As of 2001 census, total population of Khunti is 43, 4819, comprising 72.39% S.T and 5.04% S.C population.

MATERIALS AND METHODS

The authors have conducted an extensive field survey in the tribal belts and other remote villages adjoining forest areas in the district to collect ethno botanical lore. The present survey was carried out incessantly for 3 years from 2011 to 2013. Selections of villages were done on the basis of remoteness and the ethnic diversity. About 30 villages were identified for survey and documentation. Information related to ethnomedicinal plants have been documented through readymade questionnaire and personal dialogue with herbalist, vaidhyas, kabiraj, pahan, midwives, local knowledgeable persons, patients or villagers. Most of the remote villages were surveyed two times while those in the more accessible areas were visited as many as five times with an aim to obtain authentic information and collect plant specimens in their appropriate growing seasons. All wild and cultivated species were considered for this study. Information was collected concerning the parts used, mode of preparation of drug, mode of its administration, storage and the doses.

Collected plants were photographed and identified with their botanical names based on local names. For correct information and verification several books on Medicinal Plants were referred.[5-9]Documentation of information and collection and preservation of life form as herbaria was done following the methods of Jain and Rao.[10] Herbaria are kept in the Department of Botany, Ranchi University. Plants were photographed *in situ* and herbaria were also photographed.

Tools used for routine visit and collections of data are; camera, spade, knife, scissor, blades, scalpel, magnifying glasses register, diary, questionnaire, map of the study area.

RESULT AND DISCUSSIONS

As many as 90 plant species have been collected during this course of investigation but the present paper accounts for 40 plant species only, belonging to 27 families, to cure about 28 human ailments. Most of the plants

species are from families Fabaceae, Cucurbitaceae, Malvaceae, Zingiberaceae, Rubiaceae. The paper presents a brief account of the uses of various ethno medicinal plants, their parts used, method of preparation and dose prescribed (Table 1).

In most of the cases, roots were used predominantly for drug preparation followed by leaves , barks , seeds, rhizomes and in some cases flowers were also utilized. Generally, fresh plant parts were preferred than dried ones for the preparation of drug. For the drug formulations, paste, juice, infusion, decoction, powder or tablets are commonly used. For treatment either single plant or two or more plant parts are mixed as per the expertise of the herbalist. Different types of diseases treated by herbal drugs are skin diseases, epilepsy and asthma, for healing wound, to treat jaundice, diabetes, toothache, piles, gonorrhea, snake bite, scorpion sting, rheumatism, dysentery, malaria, liver disorder, eczema, ulcer, stone in bladder and several other ailments.

In recent times with the increased knowledge of life and culture of the tribal communities, the social scientists are taking interest in ethno-medicinal studies. Many works have been reported especially from among the rural and tribal communities of India and also other countries of the world. [11-18] Similar, studies have been done by some earlier workers in different areas of the Jharkhand state. [19-21]

In the course of survey interesting fact was revealed that local herbal practitioners use plants in various combinations to successfully treat several ailments. Leaves of *Acalypha indica* and *Curcuma longa* crushed into paste are applied on the affected skin to cures skin diseases; whereas root, fruit and mustard seeds crushed and mixed with karanj oil in slow fire to make paste, which is applied on eczema. It was also found that individual single plant is used to treat many ailments. For example, leaf juice of *Bacopa monnieri* is used for the treatment of epilepsy and asthma, whereas dried leaf powder is used for the treatment of nervous breakdown. *Lagenaria siceraria* root and fruit are used in jaundice, dysentery and wounds. *Ocimum sanctum* root and leaves are used as stomachache, malaria, liver disorder, antidote for snake bite scorpion sting and chronic fever.

CONCLUSION

Ethno medicinal studies are very important in order to understand the social, cultural and economic factors influencing ideas and actions concerning health and illness and also to get information on types of diseases and health problems prevalent among the people of a particular locality. Such studies may help to provide the basic health-care services to the greater part of the rural resources poor population in an effective way, provided that such studies are conducted hand-in-hand with phytochemical, pharmacological and perhaps clinical studies. In order to prevent overexploitation that could lead to extinction, efforts should be made to conserve natural resources and to domesticate selected plant species which are commonly used among the herbal practitioners. Preference for their use may be related to their availability or multipurpose use. A high level of pharmacological content should be ensured, using modern cultivation and preparation methods. The sustainable cultivation of medicinal herbs could facilitate industrial scale processing.

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Table 1. List of ethnomedicinal plants of Khunti District with local names, family and medicinal uses

Sl.	Botanical Name	Local Name	Family	Medicinal use
1.	*Abrus precatorius* L.	Ratti	Fabaceae	Leaves used for the treatment of painful swellings. Leaves chewed to relieve hoarseness of the throat.
2.	*Abutilon indicum,* L.	Kanghi	Malvaceae	Leaf, bark and root are used in ulcer, stone in bladder and diarrhea.
3.	*Acalypha indica* L.	Kuppi	Euphorbiaceae	Leaf decoction is used as laxative. It has the efficacy of loosening the intestine.
4.	*Achyranthus aspera* L.	Chirchita	Amaranthaecae	Roots chewed as tooth brush. Gives relief in toothache. Root decoction is used against diarrhea.
5.	*Aconitum heterophyllum* Wall.	Atis	Ranunculaceae	Root used in diabetes.
6.	*Acorus calamus* L.	Ghora botch	Araceae	Boiled Rhizome cures cough and also gives relief from sore throat. Rhizoid powder kills the intestinal worms.
7.	*Achyranthes aspera* L.	Latjira	Amaranthceae	Root is contraceptive and used to induce labor pain
8.	*Adina cordifolia,* Hook.	Karam	Rubiaceae	Bark and root is used in worm and wound.
9.	*Anthocephalus chinesis (Lam.)* Rich.	Kadam	Rubiaceae	Leaves and bark are used in cough, cold and weakness.
10.	*Bacopa monnieri* L.	Bramhi	Scrophulariaceae	Leaf juice is used for the treatment of epilepsy and asthma. Dried leaf powder is used for the treatment of Asthenia (Nervous breakdown).
11.	*Bombax ceiba* L.	Semal	Bombacaceae	Boiled flowers are given in Blood dysentery. Bark paste applied on the Vagina to facilitate delivery. Juice of the bark is used against rheumatism.
12.	*Brassica comprestris,* L.	Sarson	Brassicaceae	Leaf seed and root used in rheumatism, fever, body pain, cold and wound.
13.	*Butea monosperma,* Lamk.	Palas	Fabaceae	Bark and resin used in diarrhea.
14.	*Cajanus cajan* L.	Arhar	Fabaceae	New leaves are chewed in tounge wound. Seeds are edible.
15.	*Celastrus paniculatus* Wild.	Malkamni	Celastraceae	Seed oil is used for the treatment of rheumatic pains and is also used against leprosy.
16.	*Cucumis sativus* L.	Khira	Cucurbitaceae	Used in hypoglycemic blood sugar.
17.	*Curcuma longa* L.	Haldi	Zingiberaceae	Rhizomes used as antiseptic, analgesic and cures diabetes .
18.	*Cynodon dactylon* L.	Doob	Poaceae	Leaves paste and juice is used in cheek bleeding from cut, Nose trouble.
19.	*Datura metel* L.	Dhatura	Solanaceae	Root, leaf and seed is used in swelling & pain.
20.	*Ficus religiosa* L.	Peepal	Moraceae	Root and fruit are used in gonorrhea, wounds and constipation.

21.	*Hibiscus cannabinus*, L.	Patsen	Malvaceae	Leaves and fruits used in dysentery, cut and wounds.
22.	*Lagenaria siceraria* (Mol.) Standl.	Kaddu /Lauki	Cucurbitaceae	Root, and fruit used in jaundice, dysentery and wounds.
23.	*Linum usitatisimum* L.	Alsi	Linaceae	Root, fruit and mustard seed crushed is mixed with karanj oil in slow fire to make paste, which is applied on eczema.
24.	*Melilotus alba* Desr.	Ran Methi	Fabaceae	Seed and leaves used in diabetes, pain and wounds.
25.	*Momordidca charantica,* L.	Karela	Cucurbitaceae	Useful in blood purification and diabetes.
26.	*Musa paradisiaca* L.	Kela	Musaceae	Root, stem and leaves used in dysentery and diabetes. Fruit edible.
27.	*Nyctanthus arbortristis* L.	Harsingar	Oleaceae	Fruits useful in piles.
28.	*Ocimum sanctum* L.	Tusli	Lamiaceae	Root and leaves are used as stomachache, Malaria, Liver disorder, chronic fever, Antidote for snake bite and scorpion sting.
29.	*Rauwolfia serpentina* (L.) Benth. *ex* Kurz	Sarpgandha	Apocynaceae	Root is used against high blood pressure and mental disorder
30.	*Ricinus communis,* L.	Arandi	Euphorbiaceae	Bark, Leaves and Seeds are used as purgative and burn (castor oil), headache.
31.	*Sesamum indicum* L.	Til	Pedaliaceae	Leaves and seed used in piles, ulcer and dysentery.
32.	*Sesbania grandiflora* (L.) Pers.	Augast	Fabaceae	Bark, Leaves and flower are used in pox and headache.
33.	*Swertia angustifolia Buch.-Ham*.	Chiraita	Gentianaceae	Juice of leaves blood purifier and used in high blood sugar.
34.	*Synzygiun cumini* L.	Jamum	Myrtaceae	Fruit used in stomach pain and diabetes.
35.	*Thevetia neriifolia*, Juss.	Pilikaner	Apocynaceae	Root is used for diabetes.
36.	*Trachyspermum ammi* L.	Ajwain	Apiaceae	Used in stomach ache, constipation and gastric trouble.
37.	*Vigna mungo,* L.	Urd	Fabaceae	Fully fried seeds with mustard oil used in ear ache.
38.	*Vitex negundo* L.	Sinduar	Verbenaceae	Leaves used in headache and rheumatism.
39.	*Zizyphus mauritiana* Lamk.	Ber	Rhamnaceae	Root and bark used in wound and diarrhoea.
40.	*Zingiber officinale* Rosc.	Adarak	Zingiberaceae	Rhizome is used as anthelmintic, carminative, laxative, and stomachic.

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